Consider infinite trees over some finite alphabet A, i.e. $\mathrm{Tr}_A = \big(\{\mathtt{L},\mathtt{R}\}^* \to A\big)$. Recall that for a node $u \in \{\mathtt{L},\mathtt{R}\}^*$ and a tree $t \in \mathrm{Tr}_A$, by $t{\restriction}_u$ we denote the tree $t{\restriction}_u(v) = t(u \cdot v)$. In this case we say that t *contains* $t{\restriction}_u$ *under* u. By $u \preceq v$ we denote the prefix order (i.e. the descendant order on nodes), with the root ϵ being the minimal element.

We say that a tree $t \in \mathrm{Tr}_A$ is *constant* if for every node $u \in \{\mathtt{L},\mathtt{R}\}^*$ we have $t(u) = t(\epsilon)$, i.e. all labels in t are equal.

We say that a tree $t \in \mathrm{Tr}_A$ is *dense* if for all nodes $u, v \in \{\mathtt{L},\mathtt{R}\}^*$ there exists a node $v' \succeq v$ such that $t{\restriction}_{v'} = t{\restriction}_u$. In other words, whenever t contains some tree $t{\restriction}_u$ then it contains it under every node v.

We say that a tree $t \in \mathrm{Tr}_A$ is *superdense* if it is dense and contains infinitely many distinct trees (i.e. $\big|\{t{\restriction}_u \mid u \in \{\mathtt{L},\mathtt{R}\}^*\}\big| \geqslant \aleph_0$).

We say that a tree $t \in \mathrm{Tr}_A$ is *megadense* if for every $t' \in \mathrm{Tr}_A$ and $v \in \{\mathtt{L},\mathtt{R}\}^*$ there exists $v' \succeq v$ such that $t{\restriction}_{v'} = t'$. In other words, t contains t' under every node v.

1 Assignment 4

Assume that the alphabet A is $\{0, 1\}$.

1. Does there exist a non-empty (i.e. $L \neq \varnothing$) regular language $L \subseteq \mathrm{Tr}_A$ of infinite trees so that every tree $t \in L$ is dense but not constant?

2. Does there exist a non-empty regular language of infinite trees so that all trees in the language are superdense?

3. Does there exist a tree which is megadense?

2 Assignment 6

Assume again that the alphabet A is $\{0, 1\}$.

1. Does there exist an infinite (i.e. $|L| \geqslant \aleph_0$) regular language $L \subseteq \mathrm{Tr}_A$ of infinite trees so that every tree $t \in L$ is dense?

3 Assignment 7

Take a tree $t \in \mathrm{Tr}_A$ and an infinite branch $\pi \in \{\mathtt{L}, \mathtt{R}\}^\omega$. By $\pi\restriction_n$ we denote the first n symbols of π. Let

$$t[\pi] = t(\pi\restriction_0) \cdot t(\pi\restriction_1) \cdots \in A^\omega.$$

It is the sequence of labels of t on the consecutive nodes on π. By $\mathrm{walks}(t)$ we denote the set

$$\left\{t[\pi] \mid \pi \in \{\mathtt{L}, \mathtt{R}\}^\omega\right\} \subseteq A^\omega.$$

Assume that the alphabet A is $\{0, 1\}$.

1. Is it the case that for every $t \in \mathrm{Tr}_A$ the language $\mathrm{walks}(t) \subseteq A^\omega$ is ω-regular?

2. Is it the case that for every regular language $L \subseteq \mathrm{Tr}_A$ of infinite trees the language $\bigcup \left\{\mathrm{walks}(t) \mid t \in L\right\}$ is ω-regular?

3. Is it the case that every non-empty regular language $L \subseteq \mathrm{Tr}_A$ of infinite trees contains a tree $t \in L$ such that the language $\mathrm{walks}(t) \subseteq A^\omega$ is ω-regular?